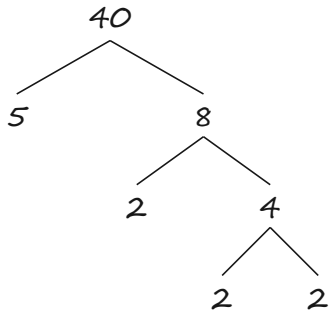
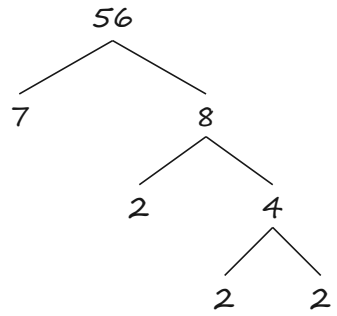
21 (a) Find the lowest common multiple (LCM) of 40 and 56





1 mark
Prime factors of both numbers

1 mark
Take the greatest power needed of each prime

$40 = 2^3 \times 5$; $56 = 2^3 \times 7$

$LCM = 2^3 \times 5 \times 7 = 280$

280

(2)

$A = 2^3 \times 3 \times 5$ $B = 2^2 \times 3 \times 5^2$

(b) Write down the highest common factor (HCF) of A and B.

Common prime factors, using their smaller powers:

$HCF = 2^2 \times 3 \times 5 = 60$

1 mark
Use only the factors present in both A and B

60

(1)

(Total for Question 21 is 3 marks)